



Instituto Infnet

TRABALHO FINAL DE CURSO

Análise de Dados no Hive

Análise dos casos de COVID-19 e SARS nos anos de 2023 e 2024.

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Introdução

Esta análise tem por motivação cruzar os dados do OpenDATASUS utilizando o Banco de Dados de Síndrome Respiratória Aguda Grave (SRAG)- incluindo dados da COVID-19 dos anos de 2023 e 2024.

Todas as bases de dados, relatórios, scripts SQL se encontra na pasta compartilhada no Google Drive conforme consta nas referências.

Objetivo

Buscar se há aumento do número de casos de COVID-19 Ano a Ano, comparar as internações em UTI entre os vacinados e não vacinados, verificar o aumento dos óbitos ano a ano para as SARS.

Fontes de Dados

A fonte de dado utilizada foi do OpenDATASUS , de onde foi feito o download das bases de dados referente aos anos de 2023 e 2024.

Os conjuntos de dados podem ser encontrados em:

<https://opendatasus.saude.gov.br/dataset/srag-2021-a-2024>

Foi realiza uma curadoria nos dados e diversas colunas do dataset original foram removidas, pois muitas não apresentavam entradas, assim como foram removidas colunas consideradas irrelevantes. Ao final ambos os dataset ficaram com 43 colunas e cerca de 250.000 (duzentos e cinquenta mil) entradas.

O **dicionário de dados** das bases após a curadoria consta da pasta compartilhada do Google Drive como **anexo A**.

Estrutura do Database no Hive

Foram criados no GCP um bucket e um cluster nó único, os arquivos CSV foram enviados ao cluster por meio do de copias pelo comando gcloud .

```
Network is unreachable) Could not connect to debian.map.fastly.net:443 (151.101.2.132), connection timed out Could not connect to debian.map.fastly.net:443 (151.101.130.132), connection timed out Could not connect to debian.map.fastly.net:443 (151.101.194.132), connection timed out Cannot initiate the connection to deb.debian.org:443 (2a04:4e42::644) . - connect (101: Network is unreachable) Cannot initiate the connection to deb.debian.org:443 (2a04:4e42:200::644) . - connect (101: Network is unreachable) Cannot initiate the connection to deb.debian.org:443 (2a04:4e42:400::644) . - connect (101: Network is unreachable) Cannot initiate the connection to deb.debian.org:443 (2a04:4e42:600::644) . - connect (101: Network is unreachable)
E: Unable to fetch some archives, maybe run apt-get update or try with --fix-missing?
gustavoqui@cluster-5f31-m:~$ gcloud
ERROR: (gcloud) Command name argument expected.
gustavoqui@cluster-5f31-m:~$ gcloud storage ls gs://projeto_inf
gs://projeto_inf/projetoHive/
gustavoqui@cluster-5f31-m:~$ gcloud storage ls gs://projeto_inf/projetoHive
gs://projeto_inf/projetoHive/
gs://projeto_inf/projetoHive/INFLUD23-26-06-2025.csv
gs://projeto_inf/projetoHive/INFLUD24-26-06-2025 (1).csv
gustavoqui@cluster-5f31-m:~$ gcloud storage cp gs://projeto_inf/projetoHive/* arquivos-projeto/
Copying gs://projeto_inf/projetoHive/INFLUD23-26-06-2025.csv to file://arquivos-projeto/INFLUD23-26-06-2025.csv
Copying gs://projeto_inf/projetoHive/INFLUD24-26-06-2025 (1).csv to file://arquivos-projeto/INFLUD24-26-06-2025 (1).csv
Completed files 2/2 | 152.2MiB/152.2MiB

Average throughput: 315.0MiB/s
gustavoqui@cluster-5f31-m:~$ cd /projeto_inf
-bash: cd: /projeto_inf: No such file or directory
gustavoqui@cluster-5f31-m:~$ ls -l
total 4
drwxr-xr-x 2 gustavoqui gustavoqui 4096 Sep  9 21:27 arquivos-projeto
gustavoqui@cluster-5f31-m:~$ cd /arquivos-projeto
-bash: cd: /arquivos-projeto: No such file or directory
gustavoqui@cluster-5f31-m:~$ cd arquivos-projeto
gustavoqui@cluster-5f31-m:~/arquivos-projeto$ ls -l
total 155820
-rw-r--r-- 1 gustavoqui gustavoqui 82709099 Sep  9 21:27 INFLUD23-26-06-2025.csv
-rw-r--r-- 1 gustavoqui gustavoqui 76840159 Sep  9 21:27 'INFLUD24-26-06-2025 (1).csv'
gustavoqui@cluster-5f31-m:~/arquivos-projeto$
```

No HDFS os arquivos foram enviados por meio de PUT . Os comandos utilizados seguem abaixo .

```
hdfs dfs -ls /
```

```
hdfs dfs -mkdir /user/gustavoqui/arquivos-projeto
```

```
hdfs dfs -mkdir /user/gustavoqui/arquivos-projeto/2023
```

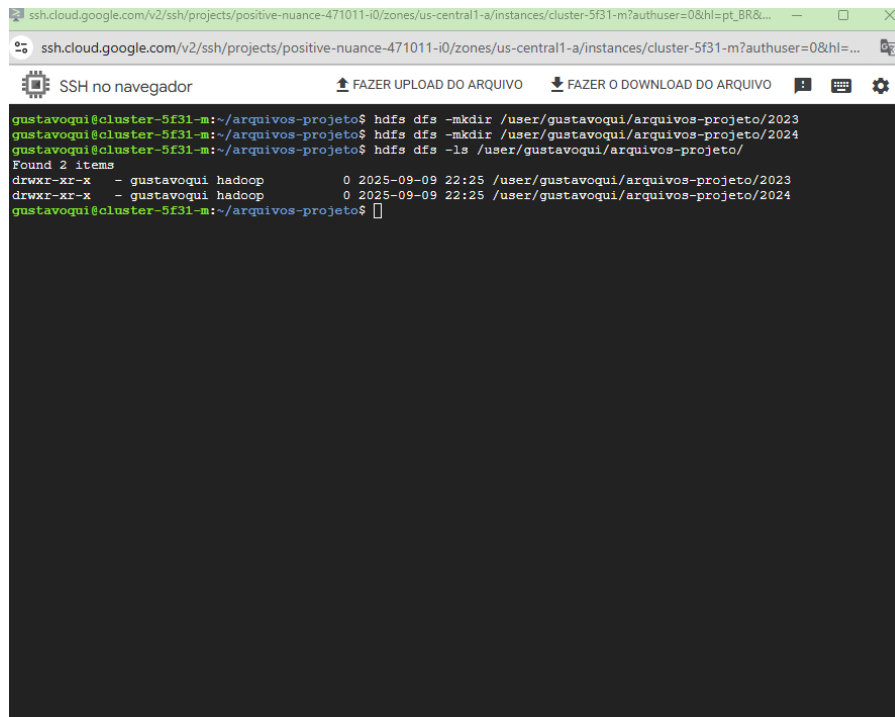
```
hdfs dfs -mkdir /user/gustavoqui/arquivos-projeto/2024
```

```
hdfs dfs -put ~/arquivos-projeto/INFLUD23-26-06-2025.csv /user/gustavoqui/arquivos-projeto/2023/
```

```
hdfs dfs -put ~/arquivos-projeto/INFLUD24-26-06-2025.csv /user/gustavoqui/arquivos-projeto/2024/
```

```
hdfs dfs -ls -R /user/gustavoqui/arquivos-projeto
```

Criação dos diretórios no HDFS



The screenshot shows a terminal window with the following commands and output:

```
gustavoqui@cluster-5f31-m:~/arquivos-projeto$ hdfs dfs -mkdir /user/gustavoqui/arquivos-projeto/2023
gustavoqui@cluster-5f31-m:~/arquivos-projeto$ hdfs dfs -mkdir /user/gustavoqui/arquivos-projeto/2024
gustavoqui@cluster-5f31-m:~/arquivos-projeto$ hdfs dfs -ls /user/gustavoqui/arquivos-projeto/
Found 2 items
drwxr-xr-x - gustavoqui hadoop          0 2025-09-09 22:25 /user/gustavoqui/arquivos-projeto/2023
drwxr-xr-x - gustavoqui hadoop          0 2025-09-09 22:25 /user/gustavoqui/arquivos-projeto/2024
gustavoqui@cluster-5f31-m:~/arquivos-projeto$
```

Ingestão dos CSV no HDFS por meio do comando PUT.

```
sshcloud.google.com/v2/ssh/projects/positive-nuance-471011-0/zones/us-central1-a/instances/cluster-5f31-m7authuser=08hl=pt_BR&projectNumber=223752874300&useAdminProxy=true - Google Chrome
sshcloud.google.com/v2/ssh/projects/positive-nuance-471011-0/zones/us-central1-a/instances/cluster-5f31-m7authuser=08hl=pt_BR&projectNumber=223752874300&useAdminProxy=true
SSH no navegador
FAZER UPLOAD DO ARQUIVO FAZER O DOWNLOAD DO ARQUIVO
gustavoqui@cluster-5f31-m7:~/arquivos-projeto$ hdfs dfs -put /arquivos-projeto/INFLUD23-26-06-2025.csv /user/gustavoqui/arquivos-projeto/2023
put: /arquivos-projeto/INFLUD23-26-06-2025.csv: No such file or directory
gustavoqui@cluster-5f31-m7:~/arquivos-projeto$ cd ..
gustavoqui@cluster-5f31-m7:~/arquivos-projeto$ hdfs dfs -put /arquivos-projeto/INFLUD23-26-06-2025.csv /user/gustavoqui/arquivos-projeto/2023
put: /arquivos-projeto/INFLUD23-26-06-2025.csv: No such file or directory
gustavoqui@cluster-5f31-m7:~/arquivos-projeto$ hdfs dfs -put /arquivos-projeto/INFLUD23-26-06-2025.csv /user/gustavoqui/arquivos-projeto/2023/
put: /arquivos-projeto/INFLUD23-26-06-2025.csv: No such file or directory
gustavoqui@cluster-5f31-m7:~/arquivos-projeto$ hdfs dfs -put ~/arquivos-projeto/INFLUD23-26-06-2025.csv /user/gustavoqui/arquivos-projeto/2023/
gustavoqui@cluster-5f31-m7:~/arquivos-projeto$ hdfs dfs -put ~/arquivos-projeto/INFLUD23-26-06-2025.csv /user/gustavoqui/arquivos-projeto/2024/
gustavoqui@cluster-5f31-m7:~/arquivos-projeto$ hdfs dfs -put ~/arquivos-projeto/INFLUD24-26-06-2025.csv /user/gustavoqui/arquivos-projeto/2024/
gustavoqui@cluster-5f31-m7:~/arquivos-projeto$
```

Verificação se os arquivos foram colocados dentro do HDFS.

```
sshcloud.google.com/v2/ssh/projects/positive-nuance-471011-0/zones/us-central1-a/instances/cluster-5f31-m7authuser=08hl=pt_BR&projectNumber=223752874300&useAdminProxy=true - Google Chrome
sshcloud.google.com/v2/ssh/projects/positive-nuance-471011-0/zones/us-central1-a/instances/cluster-5f31-m7authuser=08hl=pt_BR&projectNumber=223752874300&useAdminProxy=true
SSH no navegador
FAZER UPLOAD DO ARQUIVO FAZER O DOWNLOAD DO ARQUIVO
gustavoqui@cluster-5f31-m7:~/arquivos-projeto$ hdfs dfs -ls /user/gustavoqui/arquivos-projeto/
drwxr-xr-x - gustavoqui hadoop 0 2025-09-09 22:39 /user/gustavoqui/arquivos-projeto/2023
-rw-r--r- 1 gustavoqui hadoop 82708472 2025-09-09 22:39 /user/gustavoqui/arquivos-projeto/2023/INFLUD23-26-06-2025.csv
drwxr-xr-x - gustavoqui hadoop 0 2025-09-09 22:39 /user/gustavoqui/arquivos-projeto/2024
-rw-r--r- 1 gustavoqui hadoop 76839732 2025-09-09 22:39 /user/gustavoqui/arquivos-projeto/2024/INFLUD24-26-06-2025.csv
gustavoqui@cluster-5f31-m7:~/arquivos-projeto$
```

Criação database **tfc_infnet** no hive.

```
sshcloud.google.com/v2/ssh/projects/positive-nuance-471011-0/zones/us-central1-a/instances/cluster-5f31-m7authuser=08hl=pt_BR&projectNumber=223752874300&useAdminProxy=true - Google Chrome
sshcloud.google.com/v2/ssh/projects/positive-nuance-471011-0/zones/us-central1-a/instances/cluster-5f31-m7authuser=08hl=pt_BR&projectNumber=223752874300&useAdminProxy=true
SSH no navegador
FAZER UPLOAD DO ARQUIVO FAZER O DOWNLOAD DO ARQUIVO
gustavoqui@cluster-5f31-m7:~$ hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jarfile:/usr/lib/tes/lib/slf4j-reload4j-1.7.36.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jarfile:/usr/lib/hadoop/lib/slf4j-reload4j-1.7.36.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFactory]
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jarfile:/usr/lib/tes/lib/slf4j-reload4j-1.7.36.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jarfile:/usr/lib/hadoop/lib/slf4j-reload4j-1.7.36.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFactory]
Hive Session ID = e50a2059-c32f-4955-ae59-b8851614694a

Logging initialized using configuration in file:/etc/hive/conf.d/hive-log4j2.properties Async: true
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.hadoop.hive.common.StringInternUtils (file:/usr/lib/hive/hive-common-3.1.3.jar) to field java.net.URL.string
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.hive.common.StringInternUtils
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
Hive Session ID = bc7952e3-60d5-4d95-8283-cc247880c3be
hive> CREATE DATABASE IF NOT EXISTS tfc_infnet;
OR
Time taken: 0.59 seconds
hive> USE tfc_infnet;
OR
Time taken: 0.075 seconds
hive> show databases
> ;
OR
default
tfc_infnet
Time taken: 0.183 seconds, Fetched: 2 row(s)
hive>
```

Criação da tabela externa **ext_covid_2023** e Ingestão dos dados no hive. DROP TABLE IF EXISTS ext_covid_2023;

```
ssh.cloud.google.com/v2/ssh/projects/positive-nuance-471011-i0/zones/us-central1-a/instances/cluster-5f31-m7authuser=0&hl=pt_BR&projectNumber=223752874300&useAdminProxy=tru...
ssh.cloud.google.com/v2/ssh/projects/positive-nuance-471011-i0/zones/us-central1-a/instances/cluster-5f31-m7authuser=0&hl=pt_BR&projectNumber=223752874300&useAd...

SSH no navegador
FAZER UPLOAD DO ARQUIVO FAZER O DOWNLOAD DO ARQUIVO

> ;
OK
default
tfc_infnet
Time taken: 0.183 seconds, Fetched: 2 row(s)
hive> DROP TABLE IF EXISTS ext_covid_2023;
OK
Time taken: 0.139 seconds
hive> CREATE EXTERNAL TABLE ext_covid_2023 (
> nu_notific STRING,
> dt_notific STRING,
> sem_not INT,
> dt_sin_pri STRING,
> sq_uf_not STRING,
> co_regiona STRING,
> id_municip STRING,
> co_mun_not STRING,
> ca_sexo STRING,
> dt_nasc STRING,
> nu_idade_n INT,
> tp_idade INT,
> con_idade INT,
> sq_uf STRING,
> id_m_resi STRING,
> co_mun_resi STRING,
> vacina INT,
> antiviral INT,
> hospital INT,
> dt_interna STRING,
> sq_uf_inte STRING,
> id_m_inte STRING,
> co_mun_inte STRING,
> nu_mun_inte STRING,
> uti INT,
> amostra INT,
> dt_coleta STRING,
> tp_amostra INT,
> pct_result INT,
> clasil_fin INT,
> criterio INT,
> evolucao INT,
> dt_evolucao STRING,
> dt_encerra STRING,
> vacina_cov INT,
> dose_1_cov INT,
> dose_2_cov INT,
> dose_1_ref INT,
> dose_2_ref INT,
> fab_cov_1 STRING,
> fab_cov_2 STRING,
> fab_cov1f STRING,
> fab_cov1f2 STRING
> )
> ROW FORMAT DELIMITED
> FIELDS TERMINATED BY ','
> STORED AS TEXTFILE
> LOCATION '/user/gustavoqui/arquivos-projeto/2023';
OK
Time taken: 0.371 seconds
hive> []
```

Realizado comando Select Count (*) para verificar a criação da tabela.

```
ssh.cloud.google.com/v2/ssh/projects/positive-nuance-471011-i0/zones/us-central1-a/instances/cluster-5f31-m7authuser=0&hl=pt_BR&projectNumber=223752874300&useAdminProxy=tru...
ssh.cloud.google.com/v2/ssh/projects/positive-nuance-471011-i0/zones/us-central1-a/instances/cluster-5f31-m7authuser=0&hl=pt_BR&projectNumber=223752874300&useAd...

SSH no navegador
FAZER UPLOAD DO ARQUIVO FAZER O DOWNLOAD DO ARQUIVO

gustavoqui@cluster-5f31-m1:~$
gustavoqui@cluster-5f31-m1:~$ hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/tez/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFactory]
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/tez/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFactory]
Hive Session ID = 183fac7b-0642-4c6c-acb9-4599d85cd745

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.hadoop.hive.common.StringInternUtils (file:/usr/lib/hive/lib/hive-common-3.13.3-jar) to field java.net.URI.string
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.hive.common.StringInternUtils
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
Hive Session ID = c7a8f3cb-8e25-494e-baf8-096bc9278e16
hive> use tfc_infnet;
FAILED: SemanticException [Error 10072]: Database does not exist: tfc_infnet
hive> use tfc_infnet;
OK
Time taken: 0.053 seconds
hive> SELECT COUNT(*) AS linhas_2023 FROM ext_covid_2023;
Query ID = gustavoqui_202509235306_3cae317e-3716-4781-8785-6a59517244d6
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1757451946236_0005)

-----
VERTICES      MODE      STATUS      TOTAL      COMPLETED      RUNNING      PENDING      FAILED      KILLED
-----
Map 1 ..... container      SUCCEEDED      1            1            0            0            0            0
Reducer 2 ..... container      SUCCEEDED      1            1            0            0            0            0
-----
VERTICES: 02/02  [=====>>>] 100%  ELAPSED TIME: 6.38 s
-----
OK
278453
Time taken: 94.474 seconds, Fetched: 1 row(s)
hive>
>
```

Realizado Select simples para verificar o upload dados.

[illegible]

Criação da tabela externa `ext_covid_2024` e Ingestão dos dados no hive.

```
ssh.cloud.google.com/v2/ssh/projects/positive-nuance-471011-02/zones/us-central1-a/instances/cluster-5f31-m?authuser=0&hl=pt_BR&projectNumber=223752874300&useAdminProxy=true - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/positive-nuance-471011-02/zones/us-central1-a/instances/cluster-5f31-m?authuser=0&hl=pt_BR&projectNumber=223752874300&useAdminProxy=true

SSH no navegador

FAZER UPLOAD DO ARQUIVO
FAZER O DOWNLOAD DO ARQUIVO

01/03/2023 28/02/2023 1 NULL 5 2 1
28/04/2023 22/04/2023 1 2 1 1 1
06/01/2023 03/01/2023 1 1 2 1 2
06/01/2023 05/01/2023 1 1 5 2 2

Time taken: 0.276 seconds, Fetched: 10 row(s)
hive> DROP TABLE IF EXISTS ext_covid_2024;
OK
Time taken: 0.034 seconds
hive> CREATE EXTERNAL TABLE ext_covid_2024 (
  > na STRING,
  > dt_notific STRING,
  > sm_not INT,
  > dt_fin_pri STRING,
  > sg_uf_not STRING,
  > co_regiona STRING,
  > id_municip STRING,
  > co_mun_not STRING,
  > co_sexo STRING,
  > dt_nasc STRING,
  > nu_idade_n INT,
  > tp_idade INT,
  > cod_idade INT,
  > sg_uf STRING,
  > id_m_resi STRING,
  > co_mun_resi STRING,
  > vacina INT,
  > antiviral INT,
  > hospital INT,
  > dt_interna STRING,
  > sg_uf_inte STRING,
  > id_m_inte STRING,
  > co_mun_inte STRING,
  > nu_un_inte STRING,
  > uri INT,
  > amostra INT,
  > dt_coleta STRING,
  > tp_amostra INT,
  > pcr_resul INT,
  > classi_fin INT,
  > criterio INT,
  > evolucao INT,
  > dt_evolucao STRING,
  > dt_amostra STRING,
  > vacina_cov INT,
  > dose_1_cov INT,
  > dose_2_cov INT,
  > dose_ref INT,
  > dose_2ref INT,
  > fab_cov_1 STRING,
  > fab_cov_2 STRING,
  > fab_covrf STRING,
  > fab_covrf2 STRING
)
>
> ROW FORMAT DELIMITED
> FIELDS TERMINATED BY ','
> STORED AS TEXTFILE
> LOCATION '/user/gustavoqui/arquivos-projeto/2024';
OK
Time taken: 0.165 seconds
hive> SELECT COUNT(*) AS linhas_2024 FROM ext_covid_2024;
```

Realizado comando Select Count (*) para verificar a criação da tabela.

```
sshcloud.google.com/v2/ssh/projects/positive-nuance-471011-40/zones/us-central1-a/instances/cluster-5f31-m7authuser=0&hl=pt_BR&projectNumber=223752874300&useAdminProxy=true - Google Chrome
sshcloud.google.com/v2/ssh/projects/positive-nuance-471011-40/zones/us-central1-a/instances/cluster-5f31-m7authuser=0&hl=pt_BR&projectNumber=223752874300&useAdminProxy=true
SSH no navegador
FAZER UPLOAD DO ARQUIVO FAZER O DOWNLOAD DO ARQUIVO
> cs_sexo STRING,
> dt_baco STRING,
> mu_idade_n INT,
> tp_idade INT,
> cod_idade INT,
> sg_uf STRING,
> id_mu_real STRING,
> co_mu_res STRING,
> vacina INT,
> antiviral INT,
> hospital INT,
> dt_interna STRING,
> sg_uf_inte STRING,
> id_mu_inte STRING,
> co_mu_inte STRING,
> mm_um_inte STRING,
> uti INT,
> amostra INT,
> dt_coleta STRING,
> tp_amostra INT,
> pct_resul INT,
> class_fin INT,
> criterio INT,
> evolucao INT,
> dt_evolucao STRING,
> dt_escorra STRING,
> vacina_cov INT,
> dose_1_cov INT,
> dose_2_cov INT,
> dose_ref INT,
> dose_ref INT,
> fab_cov_1 STRING,
> fab_cov_2 STRING,
> fab_covr1 STRING,
> fab_covr2 STRING,
> )
> ROW FORMAT DELIMITED
> FIELDS TERMINATED BY ','
> STORED AS TEXTFILE
> LOCATION '/user/gustavoqui/arquivos-projeto/2024';
OK
Time taken: 0.165 seconds
hive> SELECT COUNT(*) AS linhas_2024 FROM ext_covid_2024;
Query ID = gustavoqui_20250909235927_d2c4ef32-2bad-4d1f-9283-410d56e2d174
Total jobs = 1
Launching job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1757451946236_0005)
-----
VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container  SUCCEEDED    1         1         0         0         0         0
Reducer 2 ..... container  SUCCEEDED    1         1         0         0         0         0
-----
VERTICES: 02/02 [=====>] 100% ELAPSED TIME: 10.23 s
-----
OK
267984
Time taken: 10.702 seconds, Fetched: 1 row(s)
hive> 
```

Realizado Select simples para verificar o upload dados.

```
sshcloud.google.com/v2/ssh/projects/positive-nuance-471011-40/zones/us-central1-a/instances/cluster-5f31-m7authuser=0&hl=pt_BR&projectNumber=223752874300&useAdminProxy=true - Google Chrome
sshcloud.google.com/v2/ssh/projects/positive-nuance-471011-40/zones/us-central1-a/instances/cluster-5f31-m7authuser=0&hl=pt_BR&projectNumber=223752874300&useAdminProxy=true
SSH no navegador
FAZER UPLOAD DO ARQUIVO FAZER O DOWNLOAD DO ARQUIVO
> co_mu_inte STRING,
> mm_um_inte STRING,
> uti INT,
> amostra INT,
> dt_coleta STRING,
> tp_amostra INT,
> pct_resul INT,
> class_fin INT,
> criterio INT,
> evolucao INT,
> dt_evolucao STRING,
> dt_escorra STRING,
> vacina_cov INT,
> dose_1_cov INT,
> dose_2_cov INT,
> dose_ref INT,
> dose_ref INT,
> fab_cov_1 STRING,
> fab_cov_2 STRING,
> fab_covr1 STRING,
> fab_covr2 STRING,
> )
> ROW FORMAT DELIMITED
> FIELDS TERMINATED BY ','
> STORED AS TEXTFILE
> LOCATION '/user/gustavoqui/arquivos-projeto/2024';
OK
Time taken: 0.165 seconds
hive> SELECT COUNT(*) AS linhas_2024 FROM ext_covid_2024;
Query ID = gustavoqui_20250909235927_d2c4ef32-2bad-4d1f-9283-410d56e2d174
Total jobs = 1
Launching job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1757451946236_0005)
-----
VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container  SUCCEEDED    1         1         0         0         0         0
Reducer 2 ..... container  SUCCEEDED    1         1         0         0         0         0
-----
VERTICES: 02/02 [=====>] 100% ELAPSED TIME: 10.23 s
-----
OK
267984
Time taken: 10.702 seconds, Fetched: 1 row(s)
hive> SELECT dt_notific, dt_sin_pri, hospital, uti, class_fin, evolucao, vacina_cov
> FROM ext_covid_2024 LIMIT 10;
OK
24/01/2024      22/01/2024      1         1         4         1         2
25/04/2024      19/04/2024      1         2         4         1         1
09/01/2024      08/01/2024      1         2         4         9         2
20/02/2024      19/02/2024      1      NULL         5         1         2
31/12/2023      31/12/2023      1         2         4         1         1
31/12/2023      31/12/2023      2      NULL         4         1         2
31/12/2023      31/12/2023      1         2         4         1         1
31/12/2023      31/12/2023      1         2         4         1         1
01/01/2024      01/01/2024      1         2         4         1         1
01/01/2024      01/01/2024      1         2         4         1         2
Time taken: 0.273 seconds, Fetched: 10 row(s)
hive> 
```


Consultas SQL e Análises

Consulta o número de casos de covid-19 por ano.

-- BASE para todas as consultas

WITH base AS (

```
    SELECT 2023 AS ano, dt_notific, dt_sin_pri, hospital, uti, classi_fin, evolucao,
    vacina_cov
```

```
    FROM tfc_infnet.ext_covid_2023
```

```
    UNION ALL
```

```
    SELECT 2024 AS ano, dt_notific, dt_sin_pri, hospital, uti, classi_fin, evolucao,
    vacina_cov
```

```
    FROM tfc_infnet.ext_covid_2024
```

)

--1) CASOS CONFIRMADOS por ano (COVID = classi_fin = 5)

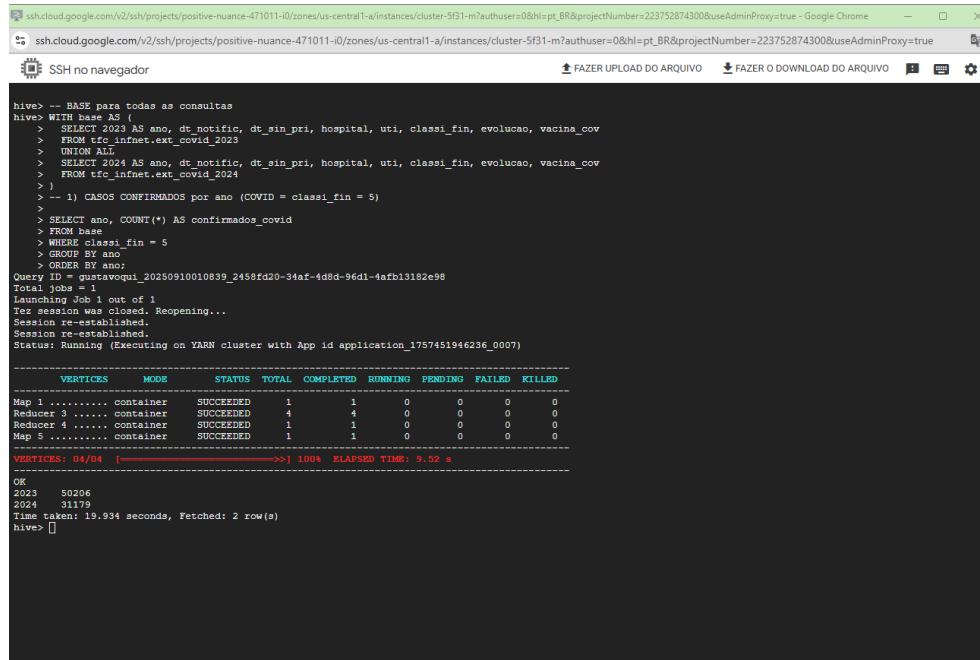
```
SELECT ano, COUNT(*) AS confirmados_covid
```

```
FROM base
```

```
WHERE classi_fin = 5
```

```
GROUP BY ano
```

```
ORDER BY ano;
```



The screenshot shows a web browser window with a terminal interface. The terminal displays the execution of Hive SQL queries. The first query is a CTE named 'base' that unions data from two tables: 'tfc_infnet.ext_covid_2023' and 'tfc_infnet.ext_covid_2024'. The second query filters for 'classi_fin = 5' and counts the number of confirmed cases ('confirmados_covid') grouped by year ('ano'). The terminal output shows the query ID, total jobs, and the execution progress of the query. A table of execution statistics is displayed, showing that all map and reduce tasks succeeded. The final output of the query is shown at the bottom of the terminal.

```
hive> -- BASE para todas as consultas
hive> WITH base AS (
>   SELECT 2023 AS ano, dt_notific, dt_sin_pri, hospital, uti, classi_fin, evolucao, vacina_cov
>   FROM tfc_infnet.ext_covid_2023
>   UNION ALL
>   SELECT 2024 AS ano, dt_notific, dt_sin_pri, hospital, uti, classi_fin, evolucao, vacina_cov
>   FROM tfc_infnet.ext_covid_2024
> )
> -- 1) CASOS CONFIRMADOS por ano (COVID = classi_fin = 5)
> SELECT ano, COUNT(*) AS confirmados_covid
> FROM base
> WHERE classi_fin = 5
> GROUP BY ano
> ORDER BY ano;
Query ID = gustavoqui_20250910010839_2458fd20-34sf-4d8d-96d1-4afb13182e98
Total jobs = 1
Launching Job 1 out of 1
Tex session was closed. Reopening...
Session re-established.
Session re-established.
Status: Running (Executing on YARN cluster with App id application_1757451946236_0007)

-----
VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container  SUCCEEDED  1      1      0      0      0      0
Reducer 3 ..... container  SUCCEEDED  4      4      0      0      0      0
Reducer 4 ..... container  SUCCEEDED  1      1      0      0      0      0
Map 5 ..... container  SUCCEEDED  1      1      0      0      0      0
-----
VERTICES: 04/04 [=====] 100% ELAPSED TIME: 9.52 s
-----
OK
2023      50206
2024      31179
Time taken: 19.934 seconds, Fetched: 2 row(s)
hive>
```

```
-- 2) INTERNAÇÕES por ano (HOSPITAL = 1) e qual ano teve mais internações.
WITH base AS (
    SELECT 2023 AS ano, dt_notific, dt_sin_pri, hospital, uti, classi_fin, evolucao,
    vacina_cov
    FROM tfc_infnet.ext_covid_2023
    UNION ALL
    SELECT 2024 AS ano, dt_notific, dt_sin_pri, hospital, uti, classi_fin, evolucao,
    vacina_cov
    FROM tfc_infnet.ext_covid_2024
)
SELECT ano,
    SUM(CASE WHEN hospital = 1 THEN 1 ELSE 0 END) AS internacoes
FROM base
GROUP BY ano
ORDER BY ano;
```

```
hives WITH base AS (
> SELECT 2023 AS ano, dt_notific, dt_sin_pri, hospital, uti, classi_fin, evolucao, vacina_cov
> FROM tfc_infnet.ext_covid_2023
> UNION ALL
> SELECT 2024 AS ano, dt_notific, dt_sin_pri, hospital, uti, classi_fin, evolucao, vacina_cov
> FROM tfc_infnet.ext_covid_2024
> )
> SELECT ano,
>        SUM(CASE WHEN hospital = 1 THEN 1 ELSE 0 END) AS internacoes
> FROM base
> GROUP BY ano;
Query ID = gwtavopai_20250910011216_2d4d13d1-4b34-479a-95e5-1649518ecc18
Total jobs = 1.
Launching job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1757451946236_0007)
```

VERTICES	MODES	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0
Reducer 3	container	SUCCEEDED	7	7	0	0	0	0
Reducer 4	container	SUCCEEDED	1	1	0	0	0	0
Map 5	container	SUCCEEDED	1	1	0	0	0	0

```
VERTICES: 04/04 (=====) 100% ELAPSED TIME: 10.43 s
OK
2023      266800
2024      256752
Time taken: 11.382 seconds, Fetched: 2 row(s)
hive>
```

```
-- 3) % UTI entre VACINADOS x NÃO VACINADOS por ano
-- (considera valores conhecidos diferentes de vacina_cov in =(1) como não vacinados e uti
in (1,2))
WITH base AS (
    SELECT 2023 AS ano, CAST(uti AS INT) AS uti, CAST(vacina_cov AS INT) AS vacina_cov
    FROM tfc_srag.ext_srag_2023
    UNION ALL
    SELECT 2024 AS ano, CAST(uti AS INT) AS uti, CAST(vacina_cov AS INT) AS vacina_cov
    FROM tfc_srag.ext_srag_2024
)
SELECT
    ano,
    CASE WHEN vacina_cov = 1 THEN 'Vacinado' ELSE 'Não vacinado' END AS status_vacinacao,
    SUM(CASE WHEN uti = 1 THEN 1 ELSE 0 END) AS uti_sim,
    COUNT(*) AS total,
    ROUND(100.0 * SUM(CASE WHEN uti = 1 THEN 1 ELSE 0 END) / COUNT(*), 2) AS perc_uti
FROM base
WHERE uti IN (1,2)
GROUP BY ano, CASE WHEN vacina_cov = 1 THEN 'Vacinado' ELSE 'Não vacinado' END
ORDER BY ano, status_vacinacao;
```

```
hive> WITH base AS (
  > SELECT 2023 AS ano, CAST(uti AS INT) AS uti, CAST(vacina_cov AS INT) AS vacina_cov
  > FROM tfc_infnet.ext_covid_2023
  > UNION ALL
  > SELECT 2024 AS ano, CAST(uti AS INT) AS uti, CAST(vacina_cov AS INT) AS vacina_cov
  > FROM tfc_infnet.ext_covid_2024
  > )
  > SELECT
  > ano,
  > CASE WHEN vacina_cov = 1 THEN 'Vacinado' ELSE 'Não vacinado' END AS status_vacinacao,
  > SUM(CASE WHEN uti = 1 THEN 1 ELSE 0 END) AS uti_sim,
  > COUNT(*) AS total,
  > ROUND(100.0 * SUM(CASE WHEN uti = 1 THEN 1 ELSE 0 END) / COUNT(*), 2) AS perc_uti
  > FROM base
  > WHERE uti IN (1,2)
  > GROUP BY ano, CASE WHEN vacina_cov = 1 THEN 'Vacinado' ELSE 'Não vacinado' END
  > ORDER BY ano, status_vacinacao;
Query ID = puravouqi_20250910018746_ad9bca08-Def2-4605-8a04-3f07997e456b
Total jobs = 1
Launching job 1 out of 1
The session was closed. Reopening...
Session re-established.
Status: Running (Executing on YARN cluster with App id application_1757451946236_0010)
```

VERTICES	MODE	STATUS	TOTAL	COMPLETED	HUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0
Reducer 3	container	SUCCEEDED	4	4	0	0	0	0
Reducer 4	container	SUCCEEDED	1	1	0	0	0	0
Map 5	container	SUCCEEDED	1	1	0	0	0	0

```
VERTICES: 04/04 [=====] 100% ELAPSED TIME: 10:02 s
OK
2023 Não vacinado 32202 112020 27.31
2023 Vacinado 42725 125339 34.09
2024 Não vacinado 36066 121409 29.71
2024 Vacinado 38912 114995 22.20
Time taken: 23.825 seconds, Fetched: 4 row(s)
hive>
```

```
-- 4) ÓBITOS por ano (EVOLUCAO = 2 normalmente indica óbito)
WITH base AS (
    SELECT 2023 AS ano, evolucao FROM tfc_infnet.ext_covid_2023
    UNION ALL
    SELECT 2024 AS ano, evolucao FROM tfc_infnet.ext_covid_2024
)
SELECT ano,
       SUM(CASE WHEN evolucao = 2 THEN 1 ELSE 0 END) AS obitos
FROM base
GROUP BY ano
ORDER BY ano;
```

```
hive> -- 4) ÓBITOS por ano (EVOLUCAO = 2 normalmente indica óbito)
> WITH base AS (
>   SELECT 2023 AS ano, evolucao FROM tfc_infnet.ext_covid_2023
>   UNION ALL
>   SELECT 2024 AS ano, evolucao FROM tfc_infnet.ext_covid_2024
> )
> SELECT ano,
>        SUM(CASE WHEN evolucao = 2 THEN 1 ELSE 0 END) AS obitos
> FROM base
> GROUP BY ano;
Query ID = guatarequal_20250910013959_f41ca247-df10-4068-b651-lcf781f5098e
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1757451946236_0009)

-----
VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container  SUCCEEDED  1        1          0        0        0        0
Reducer 3 ..... container  SUCCEEDED  7        7          0        0        0        0
Reducer 4 ..... container  SUCCEEDED  1        1          0        0        0        0
Map 5 ..... container  SUCCEEDED  1        1          0        0        0        0
-----
VERTICES: 04/04 [=====] 100% ELAPSED TIME: 19.18 s
-----
OK
2023      24944
2024      20728
Time taken: 19.95 seconds, Fetched: 2 row(s)
hive>
```

Resultados

Foi verificado com a análise dos dados, que houve uma diminuição significativa de casos de COVID-19, de 2023 para o ano de 2024.

Ainda com base na análise dos dados verificou-se que houve pouca variação no número de internações por SARS entre 2023 e 2024.

Em relação a necessidade de internação em UTIs foi verificado que o número de internados que se vacinaram é maior do que o número de internados que não se vacinaram.

Proposta de Evolução

Continuação do estudo levando em consideração os anos de 2023, 2024 e 2025.

Levantamento mês a mês para verificar em quais épocas há aumento ou diminuição dos casos, assim como quais regiões sofrem maior variação no número de casos.

Referências

Bases de dados OpenDATASUS - <https://opendatasus.saude.gov.br/dataset/srag-2021-a-2024>

Dicionário de dados OpenDATASUS - https://opendatasus.saude.gov.br/dataset/srag-2021-a-2024/resource/3135ac9c-2019-4989-a893-2ed50ebd8e68?inner_span=True

Bases de dados após curadoria - https://drive.google.com/drive/folders/1JNGDUCcoqVTZ_C-8381-Qg6arerLnvYk?usp=drive_link

Dicionário de dados após curadoria - https://drive.google.com/drive/folders/1JNGDUCcoqVTZ_C-8381-Qg6arerLnvYk?usp=drive_link

Scripts SQL utilizados - https://drive.google.com/drive/folders/1JNGDUCcoqVTZ_C-8381-Qg6arerLnvYk?usp=drive_link